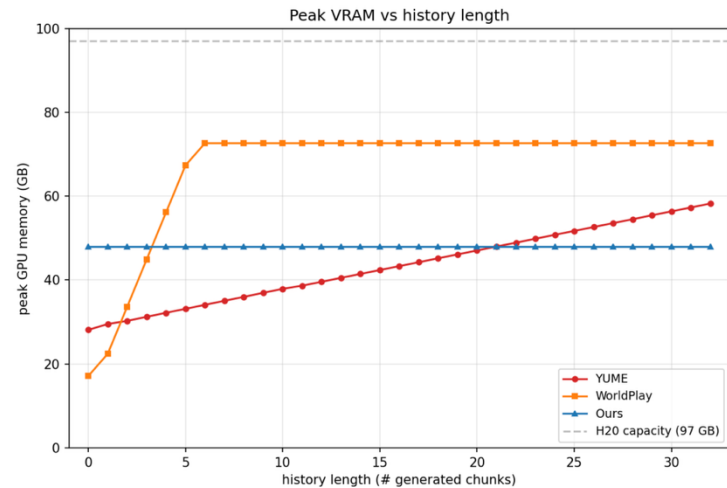
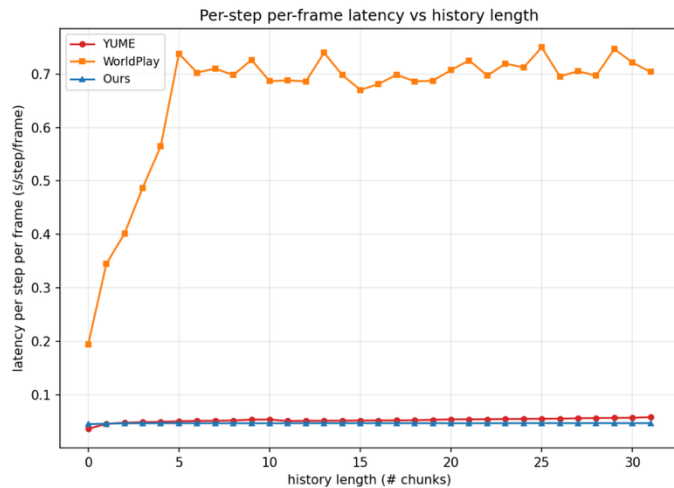
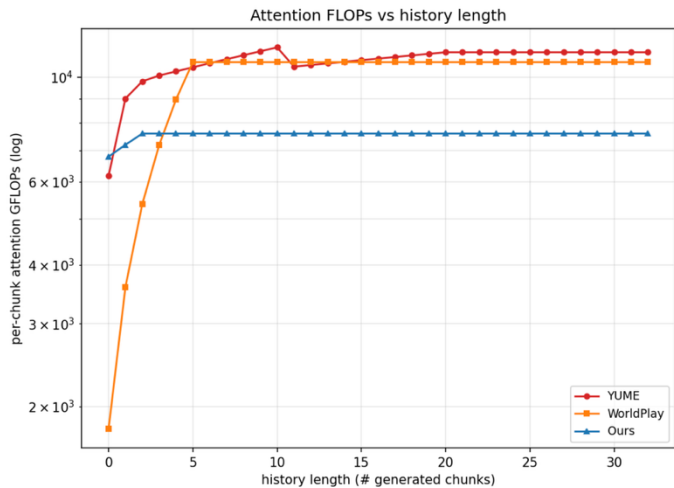




A-Fig2: More comparisons with YUME and WorldPlay, where the former uses the Framepack paradigm to compress memory to a fixed size and the latter relies on memory retrieval. Note that WorldPlay additionally sets a saturated retrieval frame count to prevent GPU out-of-memory errors caused by rising retrieval costs during long video generation. Our compressed EM memory design reaches the lowest plateau at 7.6 TFLOPs, roughly $1.4\times$ lower than WorldPlay and YUME (11 TFLOPs), and hits its performance plateau within two chunks. Latency is normalized by frames-per-chunk and sampling steps to compare mechanism overhead independent of the sampling schedule. All three methods exhibit bounded computation; our approach achieves the lowest and flattest latency (~ 0.047 s/step/frame, compared with 0.057 for YUME and 0.72 for WorldPlay). Our method maintains a constant 48 GB GPU memory footprint regardless of input history length. WorldPlay’s memory usage rises and then plateaus at 72.6 GB once its retrieval context becomes saturated. YUME’s GPU memory consumption grows approximately linearly (~ 0.95 GB per chunk). This linear growth originates from its public inference code, which caches all decoded frames in GPU memory for final video export. The dashed line denotes the 97 GB memory capacity of the H20 GPU.